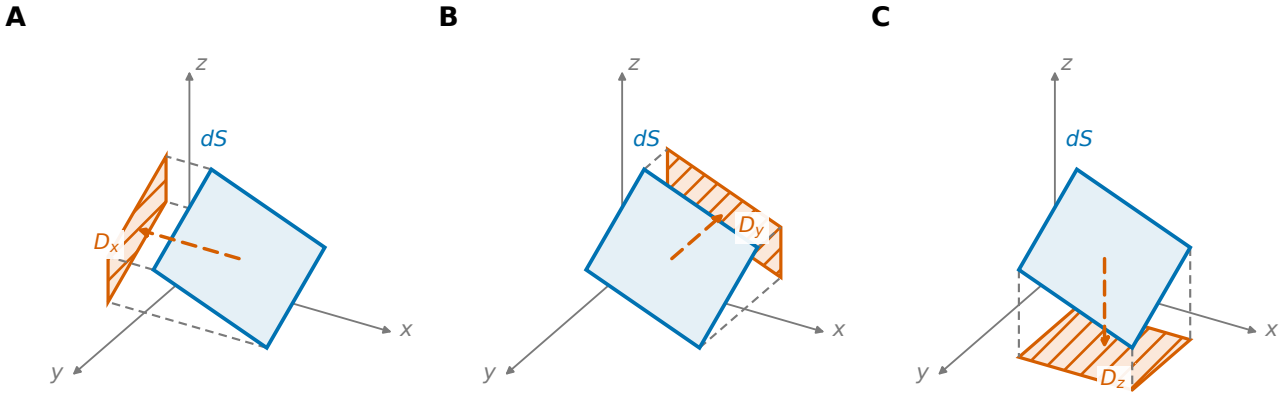

$dy\,dz = n_x\,dS$

$dz\,dx = n_y\,dS$

$dx\,dy = n_z\,dS$

$$d\mathbf{S} = (dy\,dz,\ dz\,dx,\ dx\,dy) = (n_x, n_y, n_z)\,dS \qquad (n_x, n_y, n_z > 0)$$